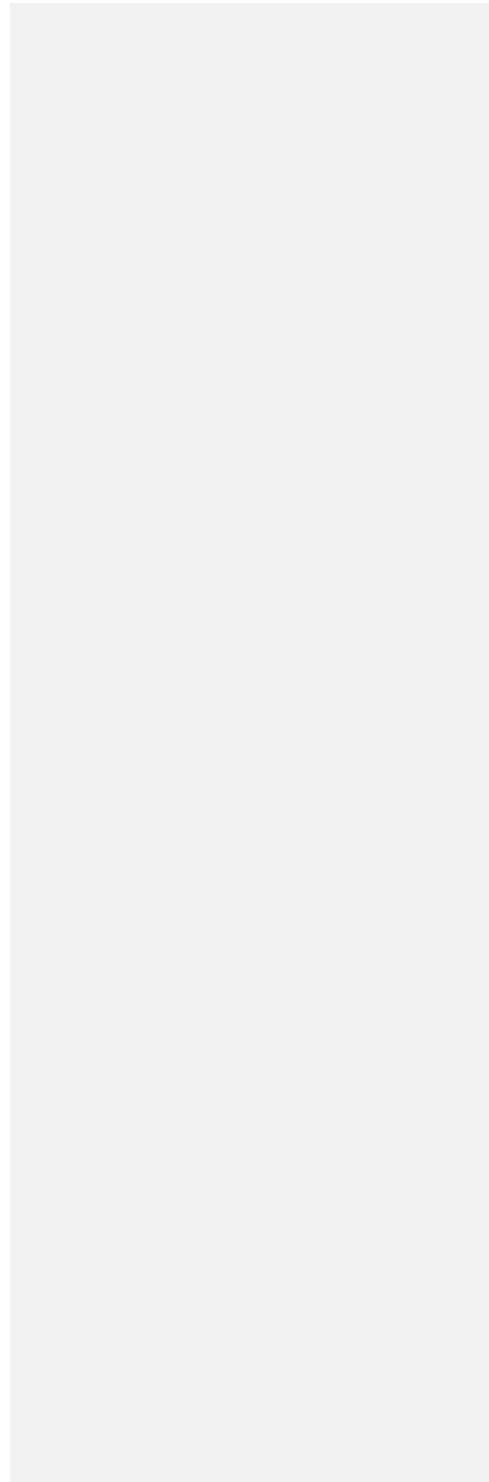


Powertrain Design Report

Formula SAE IC 2026



Abstract

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1. Introduction

1.1 Team Objectives (2026)

For the 2025-26 car, our team's objectives revolve around reliability, fit and finish and design validation (testing). For the powertrain sub-team, each of these objectives represents general objectives that are prioritized to stick to the team's philosophies this year.

1.1.1 Reliability

For reliability, the objective is to obtain better **start-up** and **idle** for **consistent performance**, improving the **drivability** for the drivers, and **eliminating leaks**.

1.1.2 Fit and Finish

The fit and finish of the powertrain system includes simple iterations on existing designs adjusting to any **issues**, **rule compliance** and potential for **increased performance**.

1.1.3 Validation and Testing

To fix on previous issues noticed in the past, this team is seeking an increase in **data acquisition** through **sensor implementation** and quality **testing plans** for validation.

1.1.4 Past Difficulties

This team has a history of shakedown time required at each testing session cutting into data acquisition time and design-based decision making based on results. There has also been an issue with our reliability on engine start-ups, idling and consistent performance and drivability.

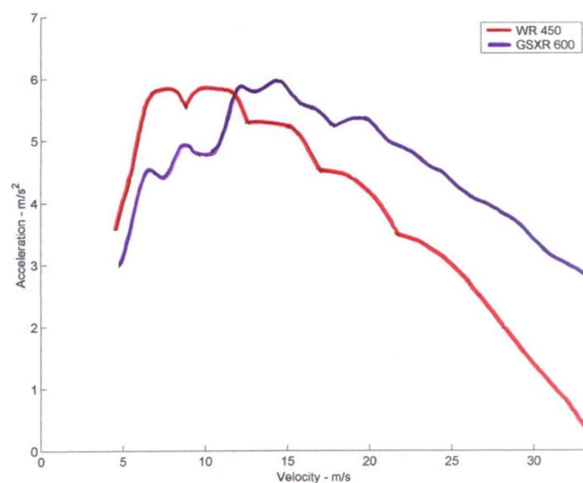
1.2 Existing Platform

1.2.1 Engine Selection

The Suzuki GSX-R600 is our engine of choice for the **availability** and **competitive** to other 4-cylinder engines. There is also a large aftermarket for this engine which has allowed and will continue to allow the implementation of performance upgrades.

Choosing a 4-cylinder engine was based around the increased power capability relevant for the acceleration event and high-speed track locations for autocross and endurance. Research shows that 4-cylinder FSAE cars see their power advantages above 45 kph due to high-end power advantages. This is demonstrated

through research and performance results in the competition with accelerated being dominated by 4-cylinders while skid-pad is dominated by 1-cylinders.



1.2.2 MoTec ECU Platform

Last year, a MoTec M150 ECU board was implemented for increased data acquisition, tune control and simplicity. Being a hobbyist platform, there is plenty of available resources and learning potential through this platform to expand the potential of the team.

1.3 Powertrain Objectives (2026)

- Iterate designs around frame changes and rule compliance
- Research and implement analysis methods
- >80 hp (increase from the 79hp from last year)
- Broad power band maximizing torque across the usable RPM range (6-14.5k RPM)
- Understanding and optimizing gearing setup
- Develop a better understanding of torque bias for optimal performance

2. Intake Manifold

2.1 Targets

- Maximizing a uniform VE for all cylinders
- Tuned for 10 000 RPM
- Throttle response and limiting choking

2.2 Design Overview

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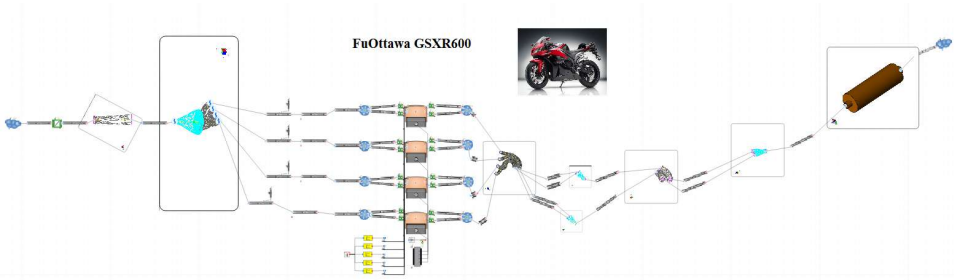


Component	Specs
General Configuration	• Square (60mm spacing)
Throttle body	• Butterfly valve • 25mm diameter
Restrictor	• 20mm diameter
Diffuser	• 3° half angle
Plenum Volume	• 3x displacement (1800cc)
Runners	• Cylinder 1-4: 197mm • Cylinder 2-3: 210mm • Elliptical Bellmouth
Material	• Nylon 12 • SLS Printed

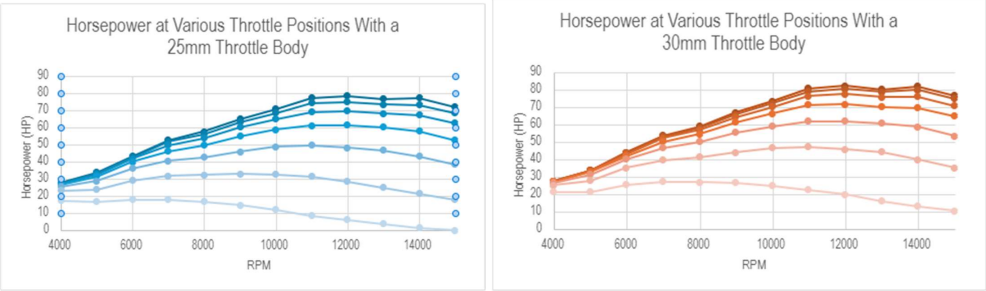
2.3 Analysis Methods

2.3.1 Throttle body

Preliminary development of a WAVE simulation:



Using this simulation, we analysed the effect of **throttle body diameters** on the **throttle pedal drive input**. The goal is to have linear correlation between your input (throttle pedal) and thee torque response at the wheels.



According to the simulation, the peak power drops from **82 hp to 79 hp** but has a more linear pedal performance. This allows driver a more consistent feel for controlling torque at the wheels.

2.3.2 Diffuser Geometry

For the diffuser design, the **pressure loss** was analyzed to understand the pressure recovery after the inevitable losses caused by the restrictor. For this, transient CFD was performed using Ansys Fluent, assuming ideal mass flow through each cylinder at 10 000 RPM.

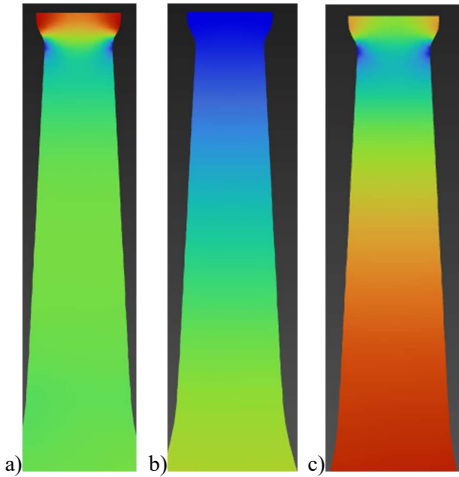
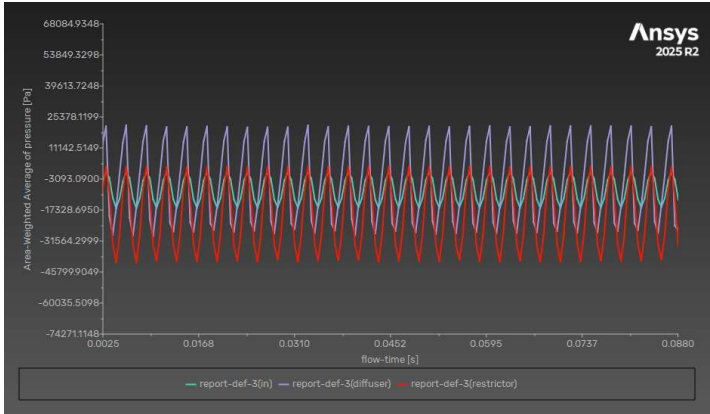
Location	Inlet	Restrictor	Mid-point	Diffuser End
Diameter	25 mm	20 mm	25 mm	30.46 mm

Mean Pressure	-8.20 kPa	-18.78 kPa	-9.39 kPa	-3.13 kPa
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From these measurements, the pressure loss at the midpoint can be calculated when the diameter reaches the same as the throttle body.

$$\Delta P_{loss} = P_{inlet} - P_{mid} = 1.2 \text{ kPa}$$

This analysis demonstrates potential for comparative studies on different expansion angles to expand our pressure recovery without creating flow separation. These analyses are done using Ansys Fluent with a transient condition at 10 000 RPM. The obtained pressure graph at the 3 measuring platforms gives the following graphs. **The mean value** is taken from the oscillating values.



Another characteristic analyzed for the diffuser is the expansion half-angle. However, the setup was not representative of the true performance of the diffuser so further work is required to validate the performance of this angle. Therefore, future work on analysis for this is required to validate the 3° expansion angle compared to the other angles recommended in literature.

2.3.3 Runner Length

Wave theory:

$$L_{Runners} = \frac{1}{2} \left(\frac{ECD \cdot a}{RPM \cdot RV} - D_{Runners} \right)$$

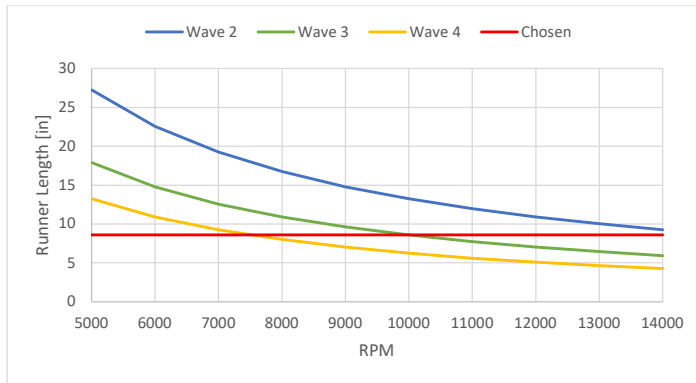
$$L_{Runners} = \frac{1}{2} \left(\frac{466^\circ \cdot 1200 ft/s}{10\,000\, RPM \cdot 3} - 1.46 in \right)$$

$$L_{Runners} = \mathbf{8.59\, in = 218\, mm}$$

Where;

- *ECD*: Effective closing duration of the intake valves
- *a*: Speed of sound through air at 60°C
- *RV*: Reflective value (harmonic number)

Given that this system was tuned for 10 000 RPM, the ideal runner lengths considered the inlet port distance to correlate to an ideal value. Cylinders **1 and 4** are 197mm long with **effective lengths of 210 mm** while the cylinders **2 and 3** are 210mm long with **effective lengths of 223mm**. Overall, this provides roughly ideal performance out of the intake runners. It is also important to analyse the performance of other RPMs for that ideal runner length.

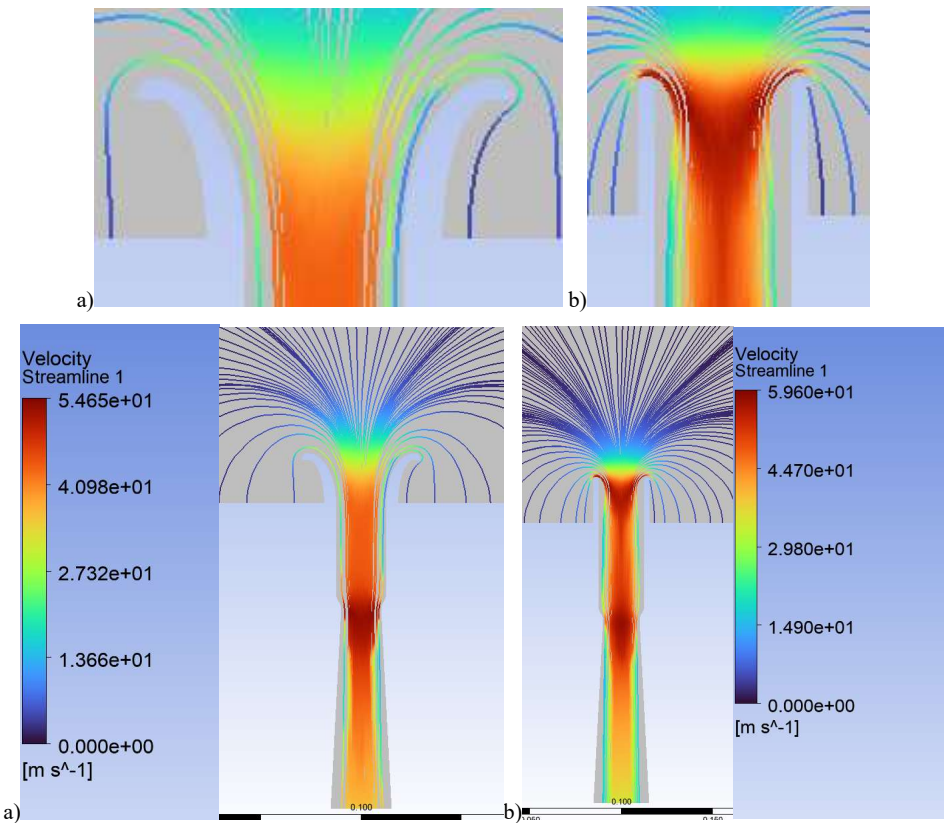


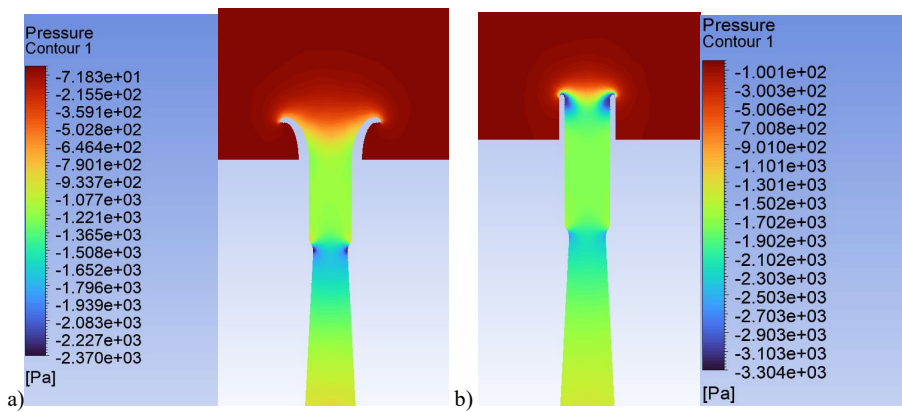
From this analysis, these runner lengths are also optimal on **second harmonic and fourth harmonic**. This can be observed at 14 000 RPM and 7 500 RPM respectively.

It is important to note with this theory that when the runner length is slightly shorter than the target, the wave is transitioning away from the valve which causes negative effects to the volumetric efficiency. In this configuration, these drawbacks happen under 7500 RPM, between 8 500 and 9 500 RPM and 12 000 RPM+.

2.3.4 Bellmouth Geometry

Following bellmouth geomtetry research, a comparison of **elliptical geometry** with straight runner ends was performed to understand the effects of the flow. The noticeable difference is the flow separation that is caused by the straight ends which chokes the flow.



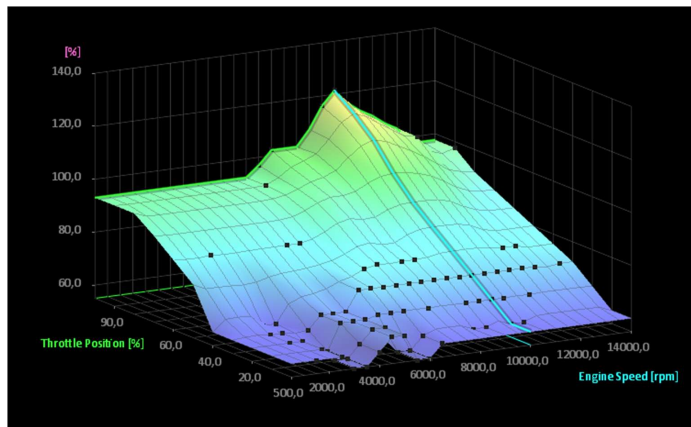


Note that the pressure scales are not the same and that the pressure at the restrictor is the same in both cases. The most notable difference is the low pressure at the inlet.

2.4 Validation

2.4.1 Dyno Data

From our dyno data, the volumetric efficiency table was produced based on the targeted lambda value of 0.85 (more information in section 5 below).



From this table, it can be concluded that the runner optimisation calculation is validated based on the spike in volumetric efficiency.

2.4.2 Throttle response

Based on the driver feedback from previous testing sessions with different intake manifolds, the 3x plenum displacement demonstrate better **feedback** and feel from the drivers. It is also important to note that this is compared to a 4x inline intake manifold. Also, this is compared to a 2x displacement plenum square intake manifold that demonstrated clear signs of choking in rapid throttle increases. This was an overcorrected design, and the plenum volume was increased accordingly.

2.4.3 EGT Data

Temperature	Cylinder 1	Cylinder 2	Cylinder 3	Cylinder 4
Idle Maximum	708.0 °C	710.7 °C	697.9 °C	746.6 °C
Idle Mean	677.5 °C	670.2 °C	652.7 °C	707.1 °C
Running Maximum	791.9 °C	842.1 °C	837.0 °C	801.6 °C
Running Mean	695.3 °C	724.7 °C	714.6 °C	701.6 °C

From this data, higher temperatures are still observed in the 2nd and 3rd cylinders demonstrating that they are getting a higher air content than the 1st and 4th cylinder. This represents uneven VE in each cylinder meaning there is more room for optimization in the system. However, a comparison with the inline intake manifold would be required to compare the improvements of the square intake manifold.

Comparing the error in VE in the cylinders, the average temperature can be used to compare the percentage of error.

$$T_{avg} = \frac{(695.3 + 724.7 + 714.6 + 701.6)}{4} = 709.1^{\circ}C$$

Running mean temperature percentage error.

Temperature	Cylinder 1	Cylinder 2	Cylinder 3	Cylinder 4
Error	-2.0%	2.2%	0.8%	-1.1%

2.5 Future Work

2.5.1 Simulation For Analysis

Optimizing the engine simulation setup would allow design iteration that can be compared with the flow bench that will together allow faster and more efficient methods for design validation. For improvements on the CFD side, there is potential for implementing **CSV files** exported from WAVE. This would allow for more

optimized flow analysis through the intake to help limit the pressure loss through the diffuser and improve bellmouth geometry.

2.5.2 Flow Bench

For validation real setup, utilizing a flow bench to compare expected flow potential through the intake and comparing different plenum sizes and bellmouth geometries. This setup is designed but not implemented or calibrated but it is in the works.

2.5.3 Testing and validation

Straight line ramp test was performed where a driver would max out the **throttle** to the preset **limiter** at 50%, 70% and 100% so that the lambda value in transient conditions can be compared between various intakes through the VE tables. However, the TPS sensor was out of sync which caused inconsistent data making the test invalid. Future tests are going to be performed to compare the three different intake manifolds.

Dyno testing would be the most optimal way to truly validate the volumetric efficiency of the intakes for a comparative study. This would also allow for proper tuning of each intake to measure the potential torque that can be obtained from this system

2.5.4 Design ideas

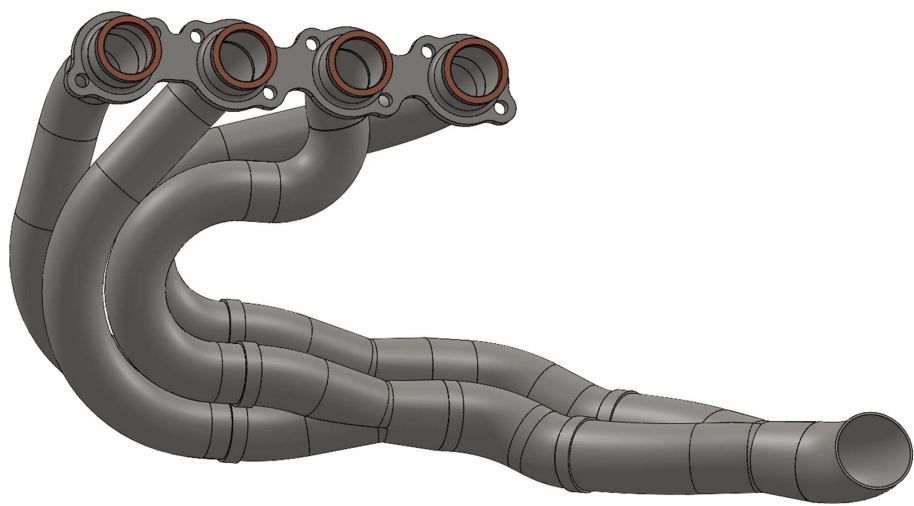
- Linear flow grid
- Remove the filter and replace with bellmouth inlet
- Extending the diffuser into the plenum
- Extending the runner in the plenum for increased packaging
- Variable runner length
- Bellmouth at the inlet
- Move the inlet away from the headrest

3. Exhaust Manifold

3.1 Targets

- Tuned for 10 000 RPM
- Optimize power across the usable RPM band
- Manufacturable design
- Better understanding of exhaust performance

3.2 Design Overview



Component	Specs
General Configuration	• 4-2-1 • Single piece header connection
Runner	• Pipe diameter: 1.5 in • Pipe thickness: 0.065 in • Cylinder 1 length: 13.9 in • Cylinder 2 length: 13.3 in

	• Cylinder 3 length: 13.6 in • Cylinder 4 length: 13.7 in
Collector length	• Diameter: 1.75 in • Length: 11 in

3.3 Analysis Methods

3.3.1 Runner Dimensioning

The first step to dimensioning the runners is to select a pipe diameter around the idea that the pulse speed is 240 ft/s (research-based method for optimal speed for scavenging). This is measured per cylinder and per cycle so 9.15cc (150cc) is one cylinder and 88 200 is the magic constant converting the ratio to an area.

$$Area = \frac{9.15 \text{ in}^3 \times 10\,000 \text{ RPM}}{88\,200} \approx 1.037 \text{ in}^2$$

$$ID = \sqrt{\frac{1.037}{0.7854}} = 1.15 \text{ in ID} \approx 1.25 \text{ in OD}$$

Since it is hard to find and expensive, **1.5 in OD tubing** has been selected to compensate. Also, the stock bike uses a 1.5 in OD exhaust. The **drawbacks** include **slower wave** propagation, **less powerful scavenging** waves and **slower gas velocity** causing higher temperature losses and lower speed of sound.

The next step is calculating the ideal runner length for a certain harmonic wave that works within the packaging restraints for the car. This is based around the exhaust valve opening duration with the estimation that the exhaust gas is at **550°C** (speed of sound 575 m/s)

$$t = \frac{\theta}{360^\circ \times (\frac{N}{60})} = \frac{266^\circ}{360^\circ \times (\frac{10\,000}{60})} = 0.00393 \text{ s}$$

$$D = t \times V_s = 0.00393 \text{ s} \times 575 \frac{\text{m}}{\text{s}} = 2.26 \text{ m}$$

For the **second harmonic**, the distance is divided into four parts making the total travel distance the following.

$$D_{2nd} = \frac{D}{4} = \frac{2.26\text{m}}{4} = 0.565\text{m} = 22.2 \text{ in}$$

Now removing the estimated head port length, collector length, flange thickness and gasket thickness, the runner length can be calculated.

$$L_{Runner} = D_{2n} - 3\text{in} - 3.5\text{in} - 1.25\text{in} - 0.22\text{in} = 14 \text{ inches}$$

Adding an acoustic correction for the location of the exhaust collector bounce back location, the acoustic runner length is calculated using a Rayleigh number of 0.6133.

$$L_{acoustic} = L_{physical} + \Delta L$$

$$\Delta L \approx 0.6133 \times r = 0.6133 \times 0.685in \approx 0.42 \text{ inches}$$

$$L_{runner} = 14 - 0.42 \approx \text{13.6 inches}$$

From these values, the exhaust design was optimized around purchasable pipe radii which is why the actual resultant runner lengths are not exactly on the target.

3.4 Validation

Refer to intake section for the **VE table** and the **EGT data**.

3.5 Future Work

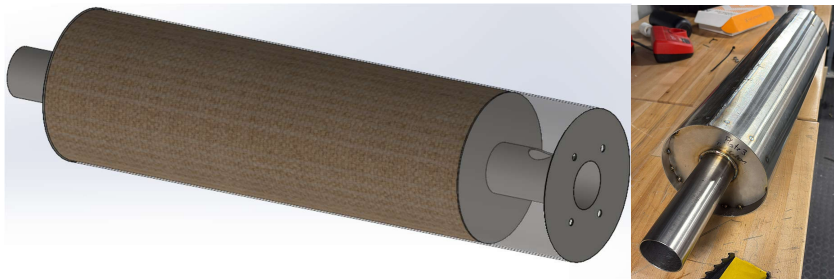
- Further research on wave theory for stepped headers
- Ceramic coating for heat retention
- Simulations: Thermal, CFD and WAVE
- Validation through dyno testing and EGT data design iterations

4. Muffler

4.1 Targets

- Transmission loss for 11 000 RPM
- Preliminary muffler design
- Hybrid configuration for broad with a slight peak

4.2 Design Overview

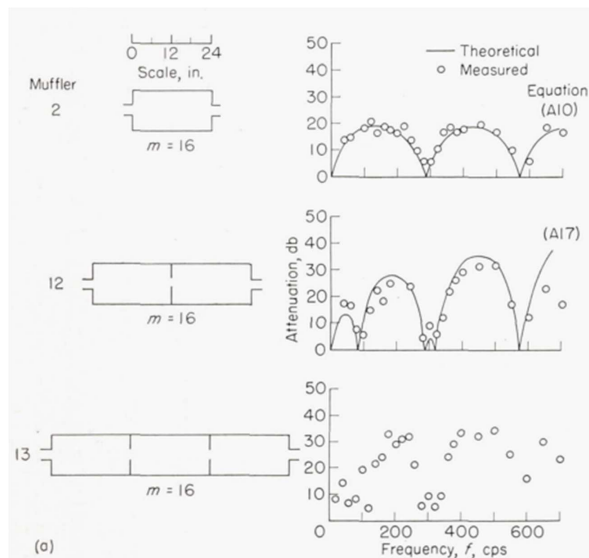


Component	Specs
General Configuration	• Hybrid muffler • Rock wool material • Helmholtz resonator
Dimensions	• Through pipe diameter: 2 in • Outer shell diameter: 6 in • Absorbative length: 19 in • Perforation diameter: • Perforation spacing: • Perforation quantity: 990 • Helmholtz length: 4 in • Helmholtz hole diameter:
Material Selection	• 16-gauge SS sheet metal

4.3 Analysis Methods

4.3.1 Chambe Count

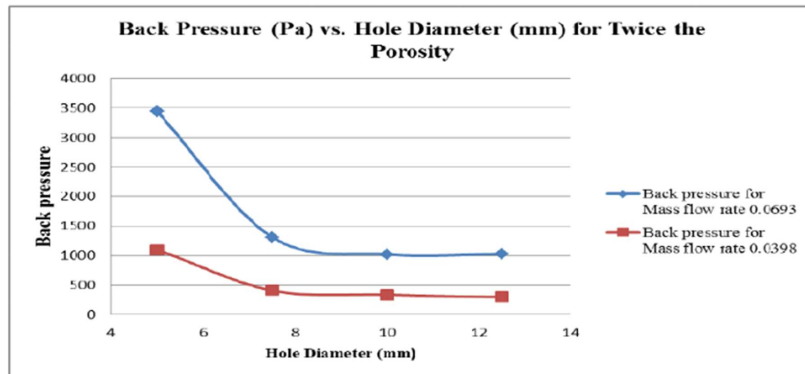
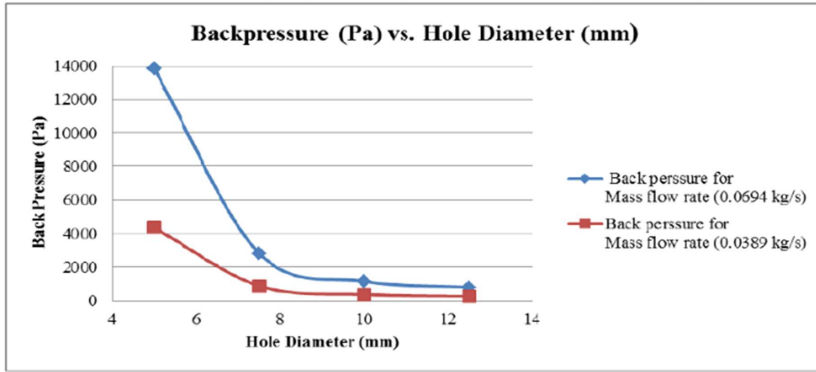
To understand the selection of a **dual expansion chamber** geometry, it is important to understand the correlation between the number of chambers and the effects on the transmission loss.



For our application, the higher RPM (higher frequency) that the engine runs with is more applicable with the two-chamber configuration where the attenuation peaks higher at each bump.

4.3.2 Absorbative Muffler

For these mufflers, the absorbative style is required to target the broad range of frequencies that are experienced from a muffler. To design this muffler, the absorptivity quality of the material and the **perforation** size and quantity are important factors for **backpressure calculations**.



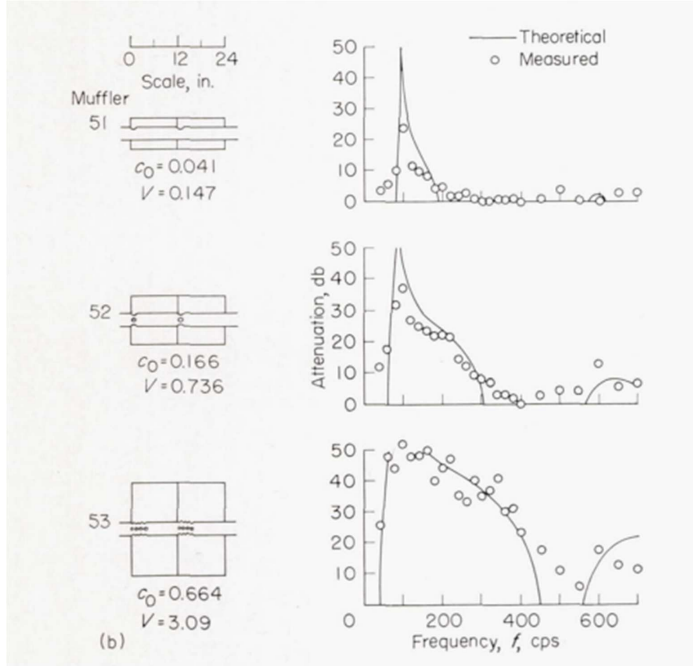
Based on the design of hole diameter, a 7.5mm hole shows limited back pressure drawback compared to larger sized holes and a higher potential density compared to larger hole sizes. Therefore, this was the select size targeting 3/8in drilled holes.

4.3.3 Reactive Muffler

The first step to analysing this muffler was to determine the frequency of the targeted RPM.

$$f = \frac{RPM \cdot N}{60 \cdot 2} = \frac{11\,000 \cdot 4}{120} = 366.6 \text{ Hz}$$

By correlating the size of the Helmholtz chamber to the expected attenuation, it can be determined that the larger it is, the broader the specific frequency effects will take place.



Working around the constraints in the car, the allowable external diameter is a chamber with 6in OD and 2in ID which gives the following volume.

$$V = \frac{\pi}{4} (OD^2 - ID^2) \cdot l = \frac{\pi}{4} (6^2 - 2^2) \cdot 4 = 100 \text{ in}^3 = 1.67 \cdot 10^{-3} \text{ m}^3$$

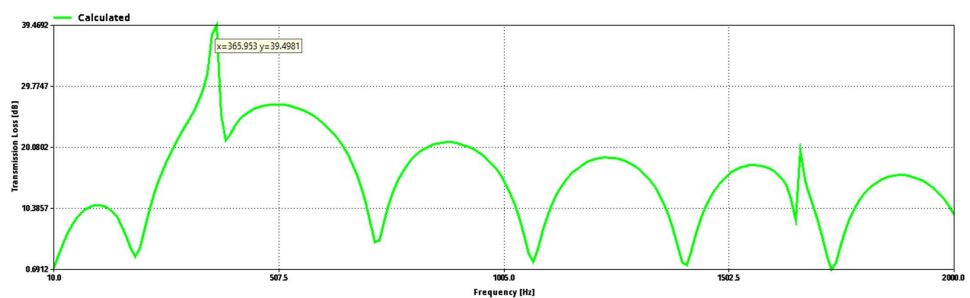
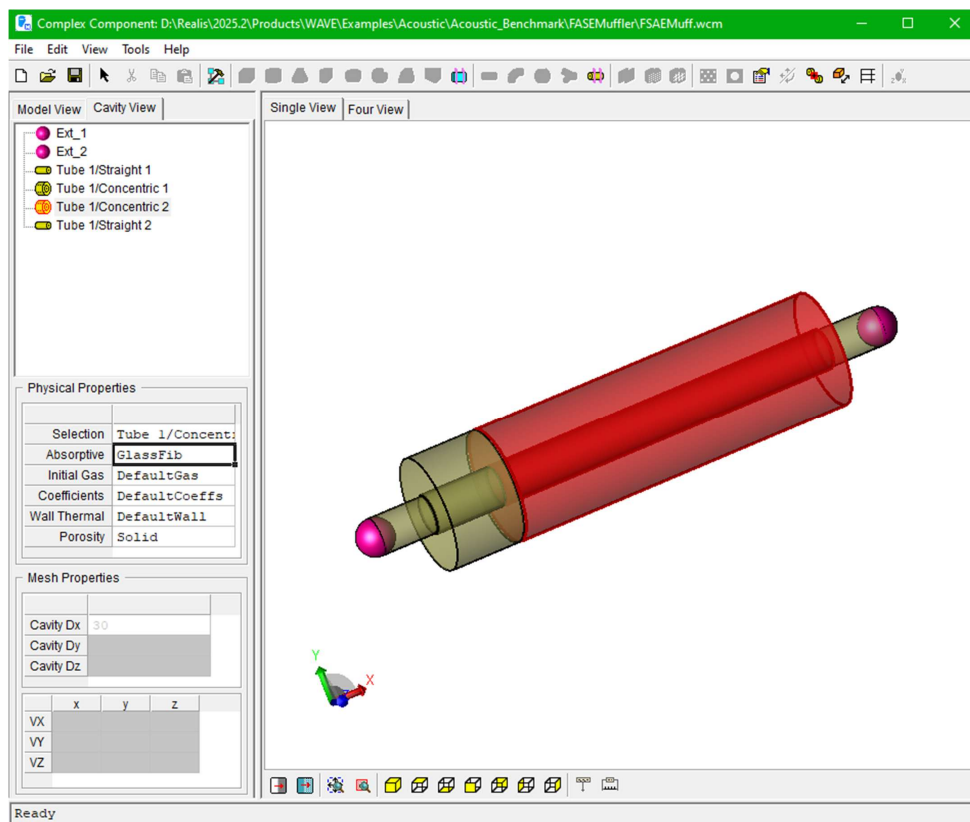
To determine the hole size of the Helmholtz resonating chamber while considering a speed of sound at 340 m/s and a resonating diameter of 1.25 in (31.75mm).

$$f_r = \frac{c}{2\pi} \sqrt{\frac{c_0}{V}}$$

$$c_0 = \left(\frac{2\pi f_r}{c} \right)^2 V = \left(\frac{2\pi(366.6)}{340} \right)^2 (1.67 \cdot 10^{-3}) = 0.077 \text{ m}$$

4.3.4 Acoustic WAVE Tuning

Building from these calculations, WAVE acoustic simulations were used to lay out the performance of the muffler across the broad frequency band. The final designed provided the following results where the peak can be found at the targeted 11 000 RPM with broad attenuation at the rest of the RPM band



From the simulations, it is determined that the Helmholtz resonator targets properly the 366.6 Hz target while also attenuating across the broad RPM band.

4.4 Validation

This has not yet been addressed for the muffler since it does not fit on the current setup of the car with the routing of the old exhaust.

4.5 Future Work

- Straight pipe testing
- Transmission loss calculations
- Comparison to stock muffler
- Material selection and weight consideration
- Venturi style muffler

5. Engine Performance and Tuning

5.1 Targets

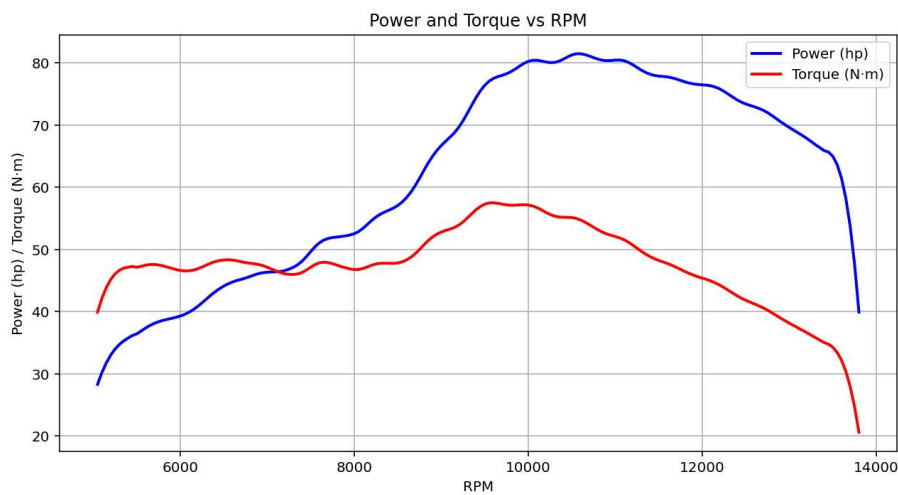
- >80hp on the dyno
- Idle, start-up and cold weather reliability
- Understanding efficiency tuning and noise tuning

5.2 Engine Specs

Component	Specs
Engine	• 2008 Suzuki GSX-R600 • Rebuilt
Customizations	• Aftermarket oil pan (2.2L capacity) • External water pump • Bosch EV14 aftermarket injectors

5.3 Performance

From the dyno session, the following torque and power values were obtained.



Peak power recoded at 82.5 hp with 58 Nm of torque.

5.4 Tuning Methods

5.4.1 Performance

Through the dyno session and during street tuning sessions, the objective is to maintain a **lambda value around 0.85-0.9**. Since our intake manifolds are restricted, the limiting factor across the RPM range is typically air quantity (volumetric efficiency). Therefore, the goal is to make use of potential power by injecting more fuel than stoichiometric potential. Also, increased fuel injection will lower the exhaust gas temperatures. This is ideal for maximizing our performance and lowering temperatures.

It was noticed during the dyno session that the performance between 0.85 and 0.9 lambda were negligible in difference. Although we don't have torque graphs

5.4.2 Efficiency

By **increasing the lambda target** with the same volumetric efficiency out of the intake, the overall fuel consumption will be reduced. The next consideration is engine temperatures and **concerns for overheating** which is when less efficient tunes with lower lambda values correlate the management of these temperatures.

5.4.3 Transient Tuning

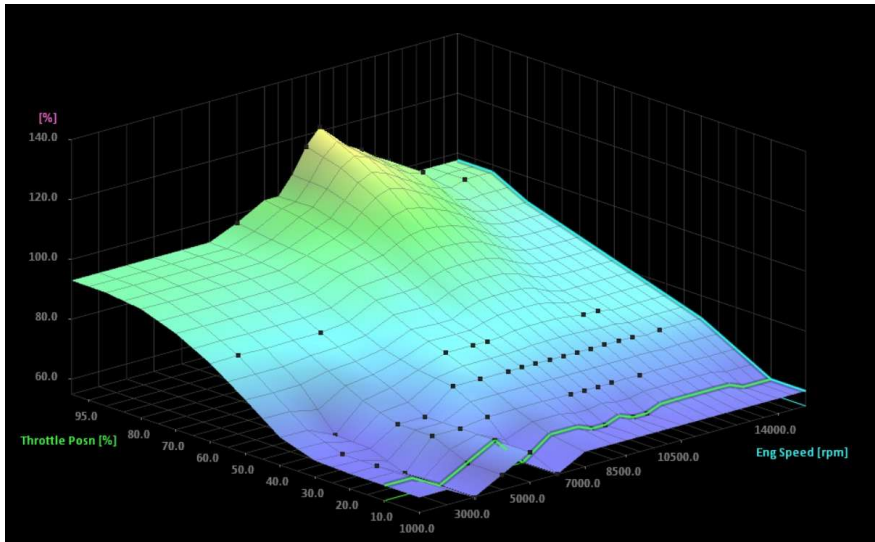
Utilizing **closed loop trims**, the MoTec platform adjusts the tune in correlation to the active values from the lambda sensor to adjust our target values in the lambda table. Also utilizing **fuel film** to correlate our intake manifold pressure with the required fuel to maintain our lambda targets during sudden transient spikes in pressure. The steady state tuning method assumes perfect conditions that are very different from drivability on the track. Therefore, utilizing this feature allows the engine to **run at our target lambda** values in varying driver conditions. This is useful for driving feel and consistent performance.

5.4.4 Cold weather startup

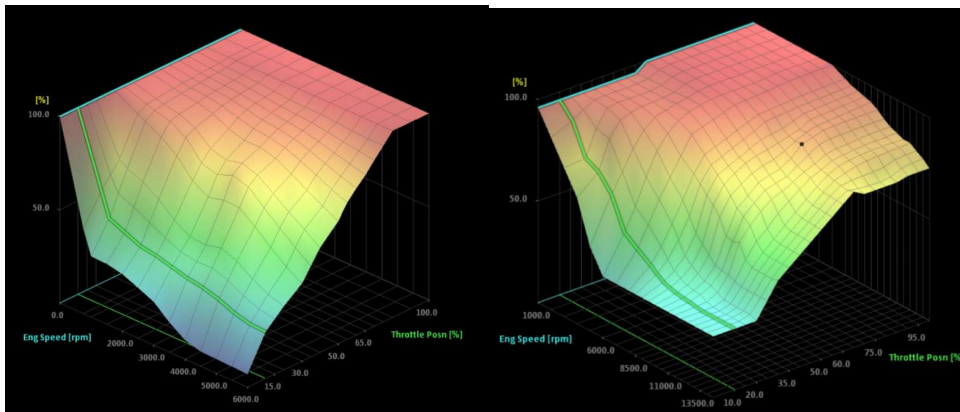
Being a Canadian team, cold weather is a persistent issue especially before competition. Multiple times throughout this year's testing session, the engine took a while to start up due to cold weather. At the dyno session, multiple hours were lost due to winter weather in the area. The solution to counteract this issue was to increase the **spark duration by a factor of 3** which increases the amount of energy to overcome the energy requirements to combust colder fuels in colder air.

5.5 Tune Graphs

Volumetric efficiency table



Improvement of the ... graph



5.6 Future Work

- Optimizing for transient tuning
- Launch control tuning

-
- Oil pressure mapping

6. Cooling

6.1 Targets

- Custom PWR Radiators for 16 kW cooling target
- Weight saving and optimization
- Car velocity 58kph

6.2 Design Overview



Component	Specs
Radiators	• Custom PWR Radiators • 9kW capabilities each
Lines	• Aluminum hard lines • AN fitting • Custom aluminum swirl-pot
Pump	• 49 LPM • 60W Centrifugal pump
Water capacity	• 3L distiller water

Velocity (m/s)	0.1	1	4	5	8	10	15
Pressure loss coefficient	396	44	16.3	14.2	10.5	9.5	8.2

6.3 Analysis Methods

6.3.1 Heating Requirements

Calculations based on recorded temperature data.

$$Q = \dot{m}c\Delta T = \left(0.49 \frac{kg}{s}\right) \left(4184 \frac{J}{kg \cdot K}\right) (8.05^\circ C) = 16kW$$

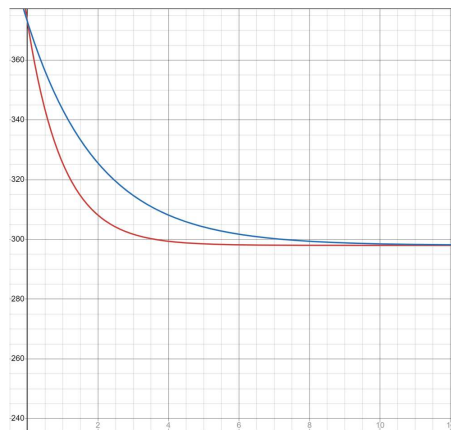
6.3.2 Series vs Parallel

In a parallel system, there is less pressure losses through the radiators due to the slower flow rate. There is also less resistance in a parallel system which means that there is a higher flow and consistency potential out of the cooling system. This more efficient cooling also allows for higher cooling potential which is most notable at higher RPM and higher power capacity flow.

$$T(t) = T_{air} + (T_{water,in} - T_{air})e^{-\frac{t}{\tau}}$$
$$\tau = \frac{\dot{m}c_p}{UA}$$

Where the flow rate in parallel is half of that in series through each radiator because the flow is split between both radiators instead of flowing through both. The effect of this reduced mass flow can be demonstrated in the graphs below where T_1 is in parallel in red and T_2 is in series in blue.

$$T_1(t) = T_{air} + (T_{water,in} - T_{air})e^{-\frac{t}{\tau}}$$
$$T_2(t) = T_{air} + (T_{water,in} - T_{air})e^{-\frac{t}{\tau/2}}$$



It can be observed that the setup in parallel has higher cooling potential by reducing the temperature more rapidly.

6.4 Validation

Data from last season demonstrates an 8°C reduction in temperature through the radiators. This data was used to calculate the heat rejection from the previous season which demonstrated consistent temperatures with minimal overheating. We will be running cooling validation tests in steady state at different speeds and RPM to measure the steady state temperature compared to the expected heat generation to calculate the heat reduction.

6.5 Future Work

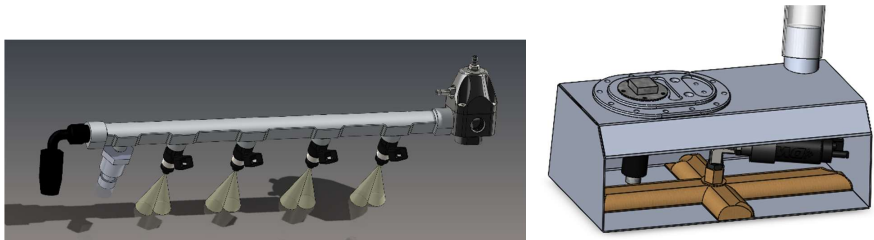
- Calculating the expected heat generation based on the combustion results
- Using power (kW) created by the engine and comparing it to complete combustion
- Calculate expected pressure loss through the lines
- Testing with and without sidepods to measure cooling capabilities

7. Fuel System

7.1 Targets

- Reduce tank volume optimized for endurance
- Weight optimization

7.2 Design Overview



Component	Specs
Fuel Type	• 93 Octane
Fuel Tank	• 8L Capacity • Aluminum
Pump System	• 100W Axial pump • Absorbative pad • Aluminum hard lines • AN fittings • 300cc/min Bosch EV14 injectors

7.3 Analysis Methods

For this sub-system, back of the napkin style calculations were performed to determine the average fuel consumption during a run at the Toronto Shootout track layout. Using fuel volume through injected quantities

$$\dot{V} = \text{Average Fuel Injection} = 155 \text{ cc/min}$$

$$V = \dot{V} \cdot t_{\text{endurance}} = 155 \frac{\text{cc}}{\text{min}} \cdot 40 \text{ min} = 6.2L$$

$$V_{\text{real}} = V \cdot SF = 8L$$

7.4 Validation

Weighing a fuel tank before and after a 2/3 endurance run, the total fuel consumption was measured at 4L. This gives a total of 6L of fuel consumption estimated in an endurance run with the maximal performance tune.

7.5 Future Work

- Testing a full endurance
- Comparing our fuel performance to other available fuels

8. Gearing

8.1 Targets

- Understanding the driveline with proficiency (transmission to wheels)
- Optimizing final drive
- Improve performance in competition acceleration event

8.2 Design Overview

Commented [SP2]: Because we theoretically chose the 4 cylinder to do well in accel

Component	Specs
General Configuration	• 6-speed sequential transmission (stock) • 1.976 primary reduction (between crankshaft and gearbox input at the clutch)
Chain Drive	• 11 tooth engine sprocket • 41 tooth rear sprocket • 520 chain (cut-to-length)
Shifting	• Multiplate wet clutch • Coil based e-shifter • Paddle shifting

Gear	Gear ratio
1 st	2.687
2 nd	2.105
3 rd	1.761
4 th	1.521
5 th	1.348
6 th	1.23

8.3 Analysis Methods

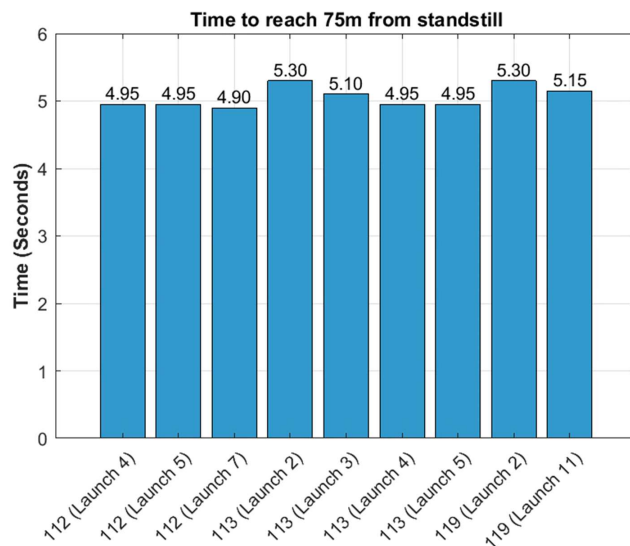
In 2025, we struggled a bit with up shifting in acceleration right before crossing the finish line, resulting in potentially lost time. In an attempt to gain some time in acceleration, this year we looked into how to elongate the gears and finish strong in acceleration at the top end of the gear. As we were running 11/41 tooth, we wanted to see what might come of increasing the drive sprocket to 12. Theoretically this is an 8% increase in gear ratios and could help maximize our shifting time to ensure we get the most out of the car.

8.4 Validation

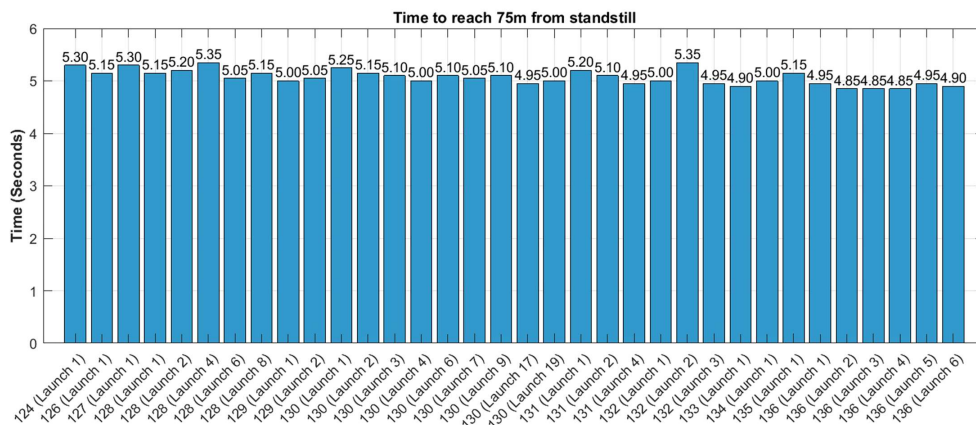
Through on-track testing of both 11 tooth and 12 tooth drive gears, we were able to determine that the 12 tooth reaches its power limit earlier than the 11-tooth does, and driver feedback suggests no difference in shifting time.

Observations:

- 11/41 ratio is proven more efficient due to longer period before power remains still
- Power limited
- Driveline ratio observed from Dyno tuning goes to 4.83/1 (for 11/41)



11 tooth sprocket accel data under 5.35s.



12 tooth sprocket accel data under 5.35s.

8.5 Future Work

- Increase in on-track testing validation through more consistent runs with the exact same conditions
 - Accel testing with same track, driver, tire pressure, etc.
- Strain gauge on driveshaft for fatigue torsion analysis
- Look into tests in the future to look into transmission choice as this was not our priority, but understanding gaps and potentially looking into custom gear box with 4 gear transmission can help improve discrepancy between gear ratios and thus longer gears

9. Limited-slip Differential

9.1 Targets

- Improve understanding of given torque bias
- Improve handling and drivability
- Understand kinematics associated with different configurations
- Redesign mounting to reflect engine changes and improve ease of chain tensioning

9.2 Design Overview



Component	Specs
General Configuration	• housing weight: 1.746kg • Two-way, dual clutch setsk • Bevel gear set • Quantity: 6cl • 25/30 Nm preload (adjustable) • 45/60 (option 2) • Engine mounted

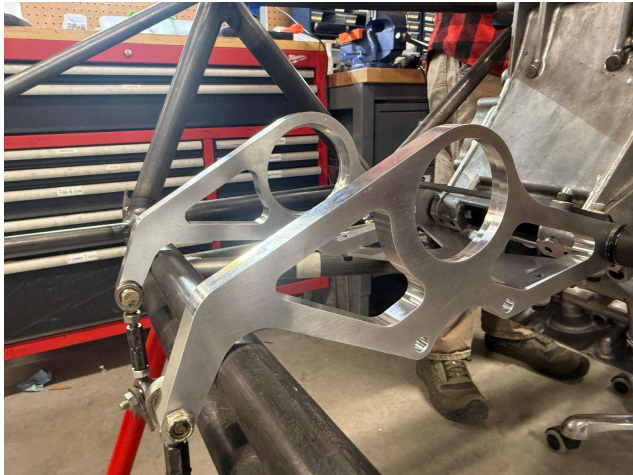
Mounts	• 7075-T6 Aluminum • ½” thick • $\pm 0.013\text{mm}$ OEM OD tolerance on bearings
Cross Bracing	• 6061-T6 Aluminum • 1/8” thick

9.2.1 Why a limited slip differential

We chose the limited slip because FSAE circuits have tight corners, hairpins, and chicanes. These features place high demand on the drivetrains ability to deliver traction from low speed on corner exit, where consequences of poor torque management are most significant for lap time. On a corner exit, lateral weight transfer moves the load to the outside rear wheel, leaving the inside rear to become light. If torque is sent to the inside wheel, much like how an open differential would, it spins and has no load to react against and will simply rotate freely in the air, resulting in wasted torque and instability. If both wheels were to rotate at the exact same speed at all times with equal amount of torque, in the manner of a spool, the car would experience severe understeer through all corners with elevated drivetrain stress. Because of this, ensuring torque is primarily sent to the outermost wheel where torque is converted to forward acceleration efficiently is optimal. The limited slip differential does exactly that, allowing speed difference that corner geometry requires while resisting that difference beyond a controlled threshold.

The specific formula student designed differential Drexler is used because of its tunability and maintainability. It houses interchangeable ramp plates with up to 6 total configurations, allowing us to change the handling characteristics for circuit specifications without the added cost of purchasing new components. Its design is intended for formula student vehicles, providing a lightweight solution at 1.746kg with chain drive integration. The Drexler has moly-coated inner splined friction plates providing a consistent and well characterised friction coefficient where its consistency is directly linked to the lockup behavior. With that, the adjustable preload and overall ease of maintenance of the Drexler has made it an optimal choice for our vehicle.

9.2.2 Mounting



9.3 Analysis Methods

9.3.1 Lockup Percentage

Lockup percentage is defined relative to a fully locked differential (100%) and fully open differential (0%). It represents the proportion of maximum possible locking torque that is being generated. The Drexler service manual provides empirical lockup values validated for the specific clutch pack geometry and friction coefficient, and these are confirmed with the following principle:

$$\text{Lockup \%} \propto \mu_{\text{clutch}} * \cot(\theta)$$

where μ_{clutch} is the friction coefficient of the molybdenum-coated plates. For a given clutch pack, mu is fixed, so lockup scales directly with $\cot(\theta)$.

Ramp angle	Theoretical lockup
30°	~88%
40°	~60%
45°	~51%
50°	~42%
60°	~29%

Of the available configurations, the ratio of 45/60 was chosen because it provides the greatest spread between entry and exit lockup values. The 22% difference between 29% and 51% means the differential behavior differs meaningfully between corner phases, where the entry is relaxed enough to allow free rotation under trail braking while still preventing oversteering. The exit is firm enough to bridge torque to the loaded outside wheel and prevent inside wheelspin, once again without understeer upon exit.

9.3.2 Torque Bias Ratio

The TBR quantifies how much torque is sent to the high traction wheel compared to the low traction wheel (valued at 1).

$$TBR = (1 + \text{lockup}\%) / (1 - \text{lockup}\%)$$

Condition	Lockup	TBR
Exit (45°)	~51%	3.08:1
Entry (60°)	~29%	1.82:1
Preload only	~7.5%	1.16:1

It should be noted that the lockup percentages present theoretical maximums locking torque that the differential can generate rather than fixed operating values. The differential modulates between zero and these limits depending on the torque input at any given moment which is in itself dependent on throttle position, engine rpm and the amount of torque flowing through the diff at that moment. Where you operate within that bound at any given corner depends on turn radius indirectly, where high speed corners can result in less time on throttle, less torque, and therefore less lockup.

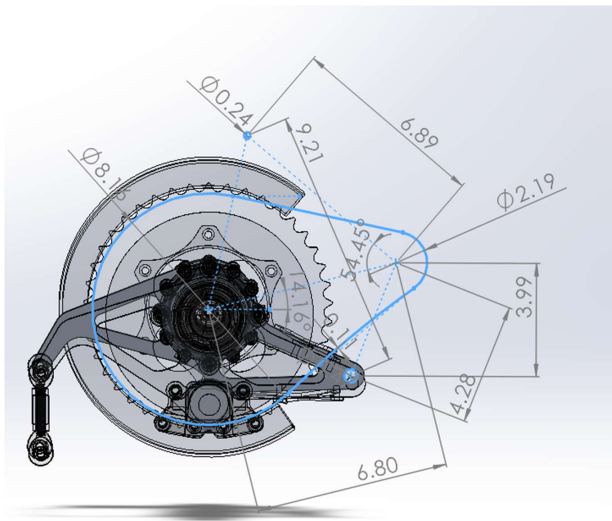
9.3.3 Understeer and oversteer implications

The resulting torque bias does theoretically favor understeer upon exit with 51% lockup and has a neutral entry, with only 29% lockup neither favoring understeer nor oversteer.

9.3.4 Mounting

As a first line of defense to properly design well-structured differential mounts, considerations were made for the one-sided chain drive including the chain line, the pull direction and the moment that it creates. The chain line directly relates to the front and rear sprockets, and in practice we use a chain sketch to define it in CAD. The chain sketch uses the sprocket pitch and geometric relations to establish constraints. The chain pull direction is the tension in the chain that acts along the chain line towards the engine sprocket. This force has both magnitude and direction relative to the diff housing and therefore the mounts need to resist this force

without allowing the diff to rotate or translate. If the mounting geometry doesn't account for this direction, the diff will deflect underload and the chain geometry changes dynamically. The resulting moment created by the chain drive acts at the sprocket which is offset from the diff's center of mass and from the mount locations. This creates a moment that tries to rotate the diff in its mounts. Therefore, understanding where your mounts sit relative to the sprocket and chain indicates whether the mount arrangement resists and effectively negates this moment. Because of this, we modelled the diff mounts from the chain sketch.



The geometry was defined by the load path directions, for 3 load cases so that the webs are loaded axially. And then optimized while ensuring they are mountable and have a shape that allows for the system to tilt up and down to get the right tensioning range. Axial loading is the design target because a web in pure axial load is working at its most efficient since the entire cross section contributes to carrying the load uniformly. A web in bending develops a stress gradient across its section, with extreme fibres doing most of the work and the center only contributing a small amount.

Once the general shape and geometry was established, the shape was refined to satisfy two constraints, ease of mounting, and chain tensioning. The geometry has to allow the component to be installed and removed from the car and the system has to be able to rotate about its mount points to vary the sprocket center distance and set the chain tension within the correct range.

Chain tensioning was designed as a turnbuckle assembly with two jam nuts on either end of a threaded rod with corresponding rod ends. These were targeted to be placed on the outside of the frame due to previous difficulties adjusting the tensioning from the inside of the frame.

Load Cases

Under these following conditions:

- engine torque (from dyno): 55.47 Nm,
- primary gear ratio: 1.976
- 2nd gear ratio (used for launching): 2.105
- 11-tooth drive sprocket
- 520 chain
- Mass: 224 kg
- peak accel: 1.32 g
- peak deceleration: 1.13g
- peak decel (coasting): 0.34g
- brake bias: 50% (at the hairpin upon corner braking at amigo)
- moment arms (from center of brake rotor to calliper): ~ 3 inches
- rolling radius: 8 inches

The load cases are:

- Acceleration at the sprocket interface along the chain line: **8298 N**
- Engine braking at the sprocket interface along the chain line: **2483 N**
- Rotor braking and at the brake caliper: **3309 N**

Calculated with (respectively):

- torque at gearbox output, drive sprocket, resulting in chain tension force.
- dyno torque, gearbox torque, chain braking force and an estimation of 30% at peak drive torque
 - This estimated 30% is an established approximation based on literature for naturally aspirated four-cylinder motorcycle engines based on their known internal friction characteristics.
- brake force, rear brake bias, tire radius and the moment arm from the rotor to the caliper.

****Load cases were done using data from the 2025 vehicle and redone using new data**

9.3.5 In board brake rotor

A custom machined cap was made for the differential in order to accommodate for the inboard rotor. Our machinist at the University helped us manufacture it to reflect the required mounting and the caliper underneath.

Why do we run an inboard brake rotor?

-
- Lighter
 - Sprung instead of unsprung on due to chassis placement instead of on wheel
 - Less rotational mass
 - One less caliper to bleed (drawback: that it's harder to bleed)
 - Keeps mass centralized
 - Easier to package wheel assemblies,

9.3.6 Understanding failure modes

In an attempt to better predict the complexity of the diff, knowing the differential's main failure modes helps establish a precedence in routine maintenance and inspection to ensure traction delivery remains consistent and that components are well kept. The main failure modes are:

- Clutch plate wear: reduction in friction coefficient and therefore reduced lockup, detected through the preload falling below 25 Nm
- Ramp face wear: change in lockup
- Bearing failure: noise
- Wrong plate order: inconsistent and unpredictable lockup
- Oil degradation: increased wear rate

Full failure prevention is not always feasible, therefore understanding what to look out for and inspect has allows us to catch wear and oil degradation early and has resulted in solution driven decisions early in the process of testing.

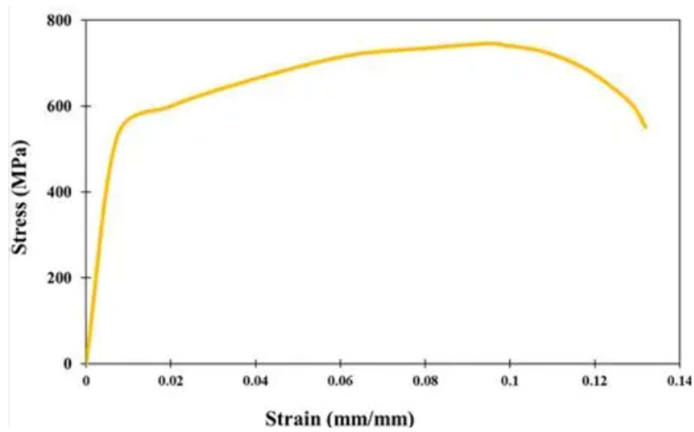
9.4 Validation

9.4.1 Torque bias testing

The current validation technique for torque bias behavior is based only on driver feedback. Other than driver feedback, the torque bias engagement itself was not physically tested alone to confirm its response.

9.4.2 Differential mounting

Finite Element Analysis was conducted for the 4 analyzed load cases on the differential unit. The diff itself was modelled as a cylinder due to its complex geometry and the run time that it would take to form a mesh and solve. Ansys was used as the selected modelling platform with 6 cores. The resulting FEA, in static structure, equivalent stress, and total deformation was compared against stress-strain graph for 7075-T6 aluminum, where yield strength 503 MPa, shear strength is 331 MPa, and fatigue strength is 159 at 5×10^8 cycles (fatigue limit).

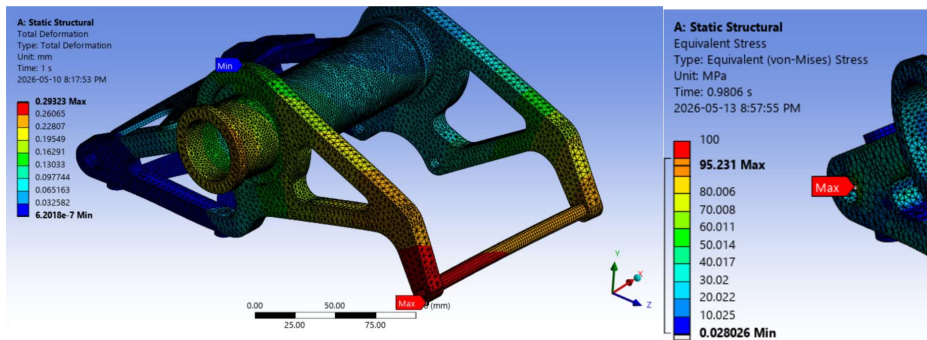
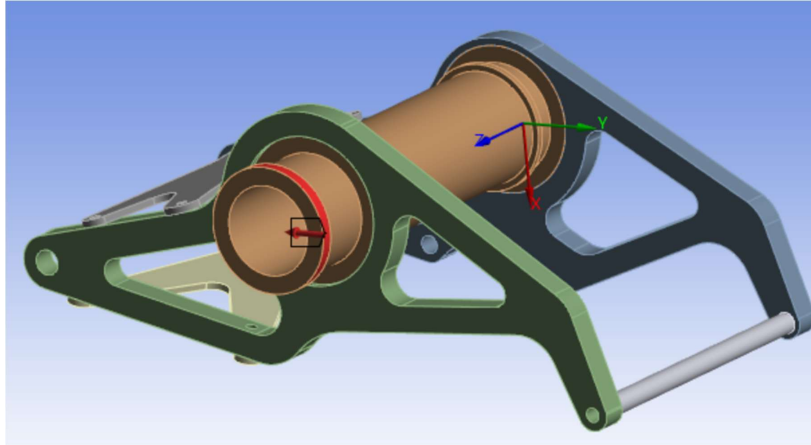


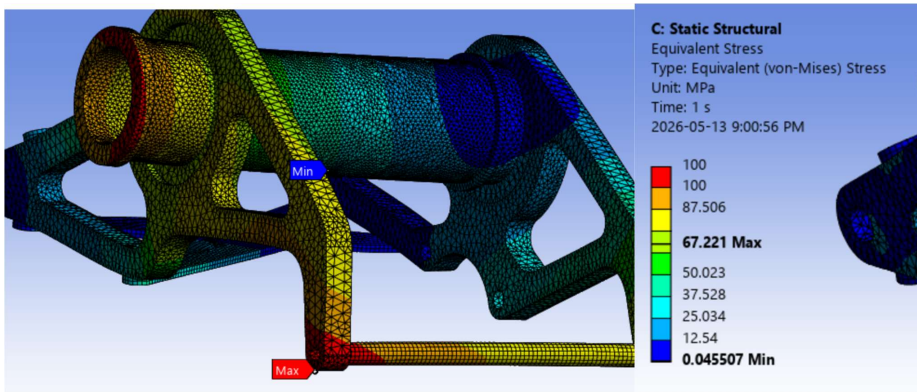
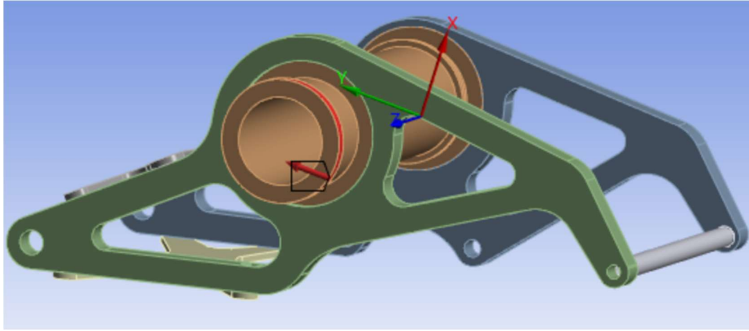
Recall the load cases are:

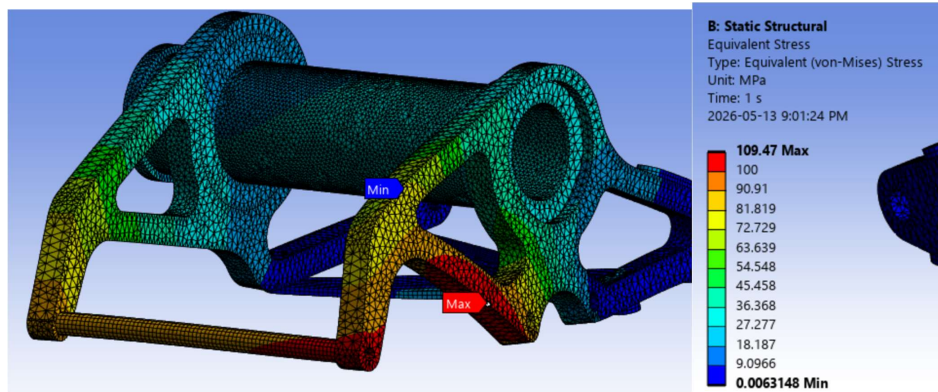
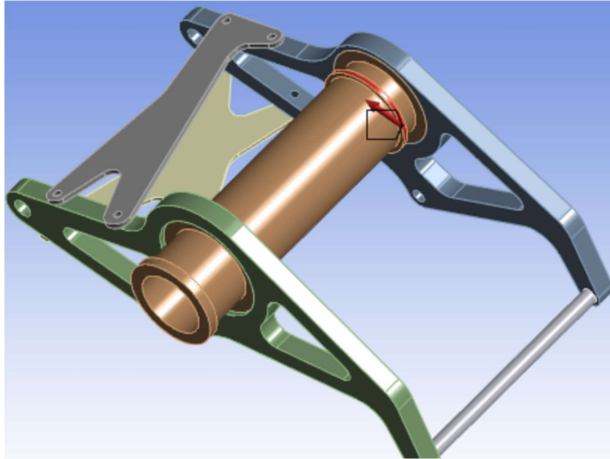
- Acceleration at the sprocket interface along the chain line
- Engine braking at the sprocket interface along the chain line
- Rotor braking and at the brake caliper

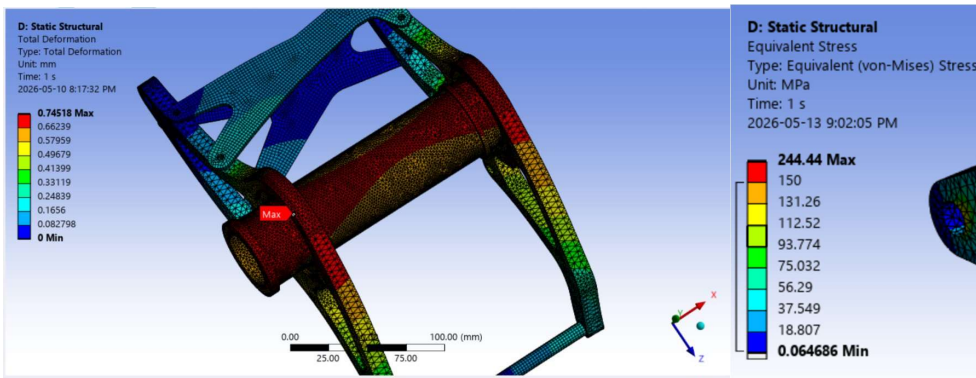
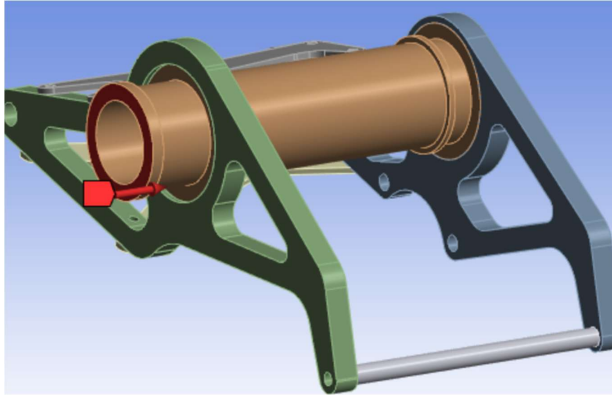
Load Case	Peak Stress [MPa] (approximate)	SF (Fatigue Limit)	SF (Yield Strength)	Total Deformation [mm]
Accel	95.23	1.67	5.28	0.29323
Rotor Braking	109.47	1.45	4.59	0.18193
Engine Braking	67.22	2.37	7.48	0.26488

Picture of the analysis described in the table follows:









9.5 Future Work

In the future, testing for the torque bias could include:

- With the diff assembled and held in a vice, applying torque to one output shaft in the drive direction and confirm resistance increases compared to the coast direction. This would validate that the two ramp faces are correctly installed and engaging as intended, eliminating failure modes.
- Upon further testing, if we notice more aggression is required, swapping the angles to 40/50 could be a solution
- Plotting yaw rate vs lateral g
- Plotting throttle position vs yaw on exit
- Plotting steering angle vs lateral g (this includes the implementation of a steering angle sensor)

-
- Improve chain tensioning system to account for lack of link worth of travel availableb